

DEPARTMENT OF MATHEMATICS
BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
MA24101: Mathematics-I (Session: MO-25)

Assignment – 4 (Module IV)

1. Evaluate the following multiple integrals:

(a) $\int_1^2 \int_y^{y^2} dx \, dy$

(c) $\int_{-1}^1 \int_{x^2}^1 \int_0^{1-y} dz \, dy \, dx$

(b) $\int_1^4 \int_0^{\sqrt{x}} \frac{3}{2} e^{\frac{y}{\sqrt{x}}} dy \, dx$

(d) $\int_0^{\frac{\pi}{4}} \int_0^{\ln(\sec v)} \int_{-\infty}^{2t} e^{2x} dx \, dt \, dv$

2. Change the order of integration of the following double integrals:

(a) $\int_0^a \int_0^x \frac{\psi(y)}{\sqrt{(a-x)(x-y)}} dy \, dx$

(b) $\int_0^{2a} \int_{\sqrt{2ax-x^2}}^{\sqrt{2ax}} f(x, y) dy \, dx$

3. Change the order of integration of the following double integrals and hence evaluate them.

(a) $\int_0^2 \int_0^{4-x^2} \frac{x e^{2y}}{4-y} dy \, dx$

(d) $\int_0^\infty \int_0^x x e^{-x^2/y} dy \, dx$

(b) $\int_0^a \int_{-\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}} dy \, dx$

(e) $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy \, dx$

(c) $\int_0^1 \int_{1-x}^{1-x^2} dy \, dx$

(f) $\int_0^\pi \int_x^\pi \frac{\sin y}{y} dy \, dx$

4. Transform from Cartesian to polar coordinates and evaluate:

(a) $\int_{-1}^1 \int_{-\sqrt{1-y^2}}^0 \frac{4\sqrt{x^2+y^2}}{1+x^2+y^2} dx \, dy$

(b) $\int_0^2 \int_0^{\sqrt{1-(x-1)^2}} \frac{x+y}{x^2+y^2} dy \, dx$

5. Find the area that lies outside the circle $r = 1$ and inside the cardioid $r = 1 + \cos \theta$.

6. Find the area of the region common to the interiors of the cardioids $r = a(1 + \cos \theta)$ and $r = a(1 - \cos \theta)$.

7. Find the volume common to the two cylinders $x^2 + y^2 = 4$ and $x^2 + z^2 = 4$.

8. Find the volume bounded by the cylinder $y^2 + 4z^2 = 16$ and the planes $x + y = 4$ and $x = 0$.

9. Find the volume of the portion of the sphere $x^2 + y^2 + z^2 = a^2$ lying inside the cylinder $x^2 + y^2 - ax = 0$.

10. Evaluate

$$\int_0^{2/3} \int_y^{2-2y} (x+2y)e^{-(x-y)} dx dy$$

using the transformation $u = x + 2y$ and $v = x - y$.

11. Evaluate

$$\iint_R e^{xy} dx dy,$$

where the region R is bounded by $y = \frac{x}{2}$, $y = x$, $xy = 1$, and $xy = 2$.

12. Find, by triple integration, the volume of the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

13. Evaluate the following triple Integral

(a) $\int_{y=0}^3 \int_{x=0}^2 \int_{z=0}^1 (x+y+z) dz dx dy$

(b) $\int_{-1}^1 \int_0^2 \int_{x-z}^{x+z} (x+y+z) dy dx dz$

(c) $\int_1^e \int_1^{\log y} \int_1^{e^x} \log z dz dx dy$

(d) $\int_{-1}^1 \int_0^{1-x} \int_0^{1-x-y} \frac{dy dx dz}{(x+y+z)^3}$

(e) $\int_0^{\pi/2} d\theta \int_0^{a \sin \theta} dr \int_1^{(a^2-r^2)/a} r dz$

(f) $\iiint_V dx dy dz$, where $V = \{(x, y, z); x^2 + y^2 + z^2 \leq a^2, 0 \leq z \leq h\}$, using cylindrical polar coordinate.

(g) $\iiint_R xy \cos yz dx dy dz$, where R is the parallelepiped $\{(x, y, z); 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq \pi/2\}$

(h) $\int_0^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-\sqrt{1-x^2-y^2}}^{\sqrt{1-x^2-y^2}} z^2 dz dx dy$ using polar coordinates